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POWERING THE FUTURE: DIGITAL TWINS FOR SMARTER, RELIABLE, AND EFFICIENT GRIDS

Presented By

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INTRODUCTION

Challenges in Present-Day Power Grid

- 📈 Rapid industrial growth increases electricity demand
- 🚗 Electrification of transport adds additional load
- 🏙️ Urbanization leads to peak load stress
- ⚠️ Existing grid struggles to maintain stability
- 🔄 Traditional grid supports only unidirectional power flow
- 🌞 Integration of renewable energy requires bidirectional flow
- 🏗️ Aging transmission & distribution infrastructure
- 🔧 High failure rate and maintenance cost
- 📉 Reduced reliability of power supply

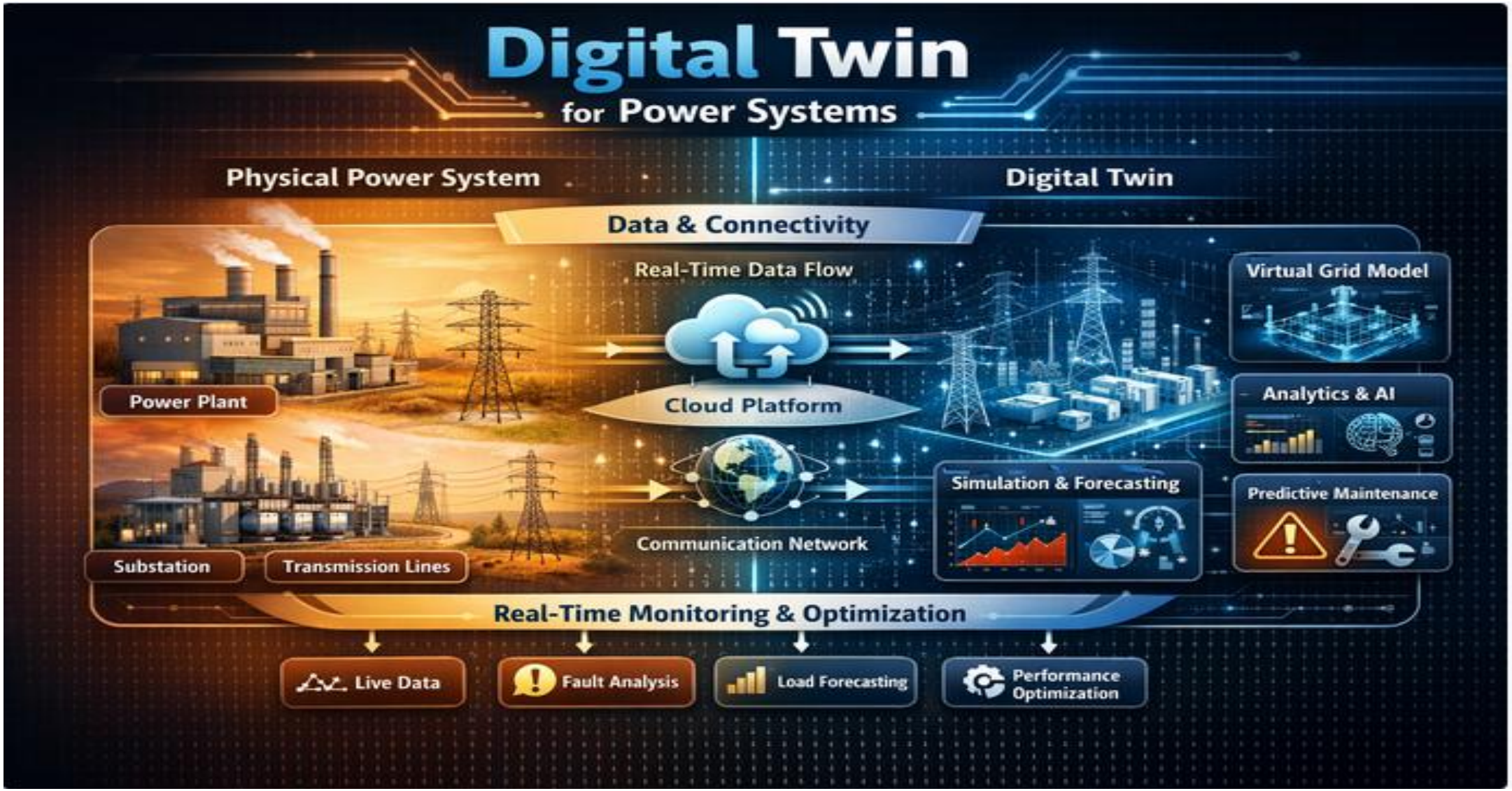
Challenges in Modern Power Grid

- 🔐 Increased cybersecurity risks (SCADA attacks, grid hacking) due to digitalization.
- 📉 High technical losses in transmission & distribution systems
- ⚡ Irregular load patterns from EV charging stations
- 📡 Need for real-time monitoring, predictive maintenance & automatic fault detection
- 🤝 Regulatory and operational challenges in peer-to-peer energy trading
- 🌍 Pressure to meet sustainability & carbon neutrality targets

SMART SOLUTIONS FOR GRID MODERNIZATION

- 🔐 Advanced cybersecurity systems (encryption, AI-based intrusion detection)
- 🧠 AI-driven Digital Twin for real-time simulation & forecasting
- 📊 Predictive maintenance & automatic fault detection
- 📡 Smart meters, PMUs, sensors & automation technologies
- 🔄 Bidirectional power flow management (V2G / G2V integration)
- 🤝 Peer-to-peer energy trading platforms for prosumers
- ⚡ Dynamic pricing & demand response mechanisms
- 🌞 Seamless integration of renewable energy & DERs
- 🌍 Strategies aligned with sustainability & carbon neutrality goals

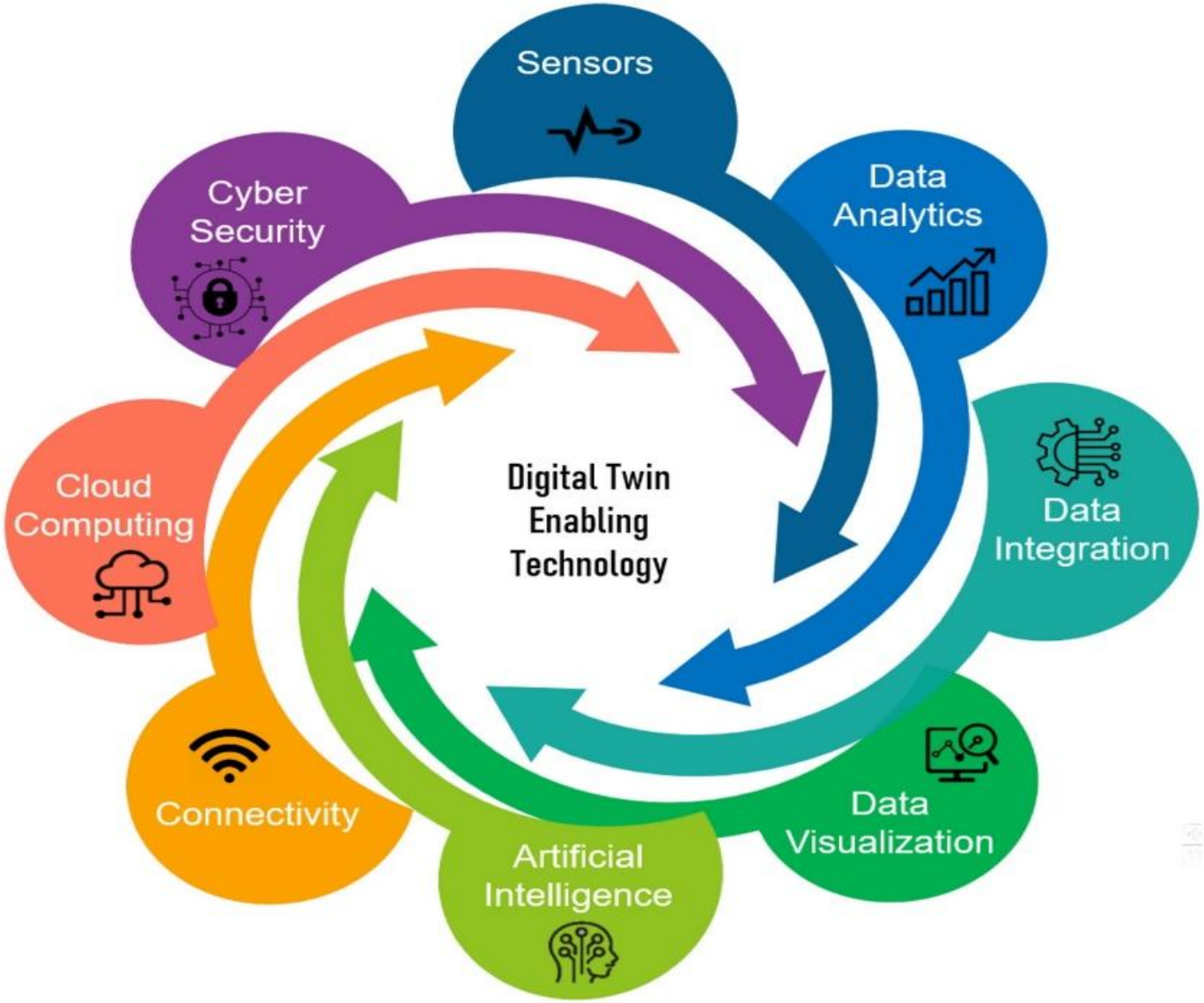
DIGITAL TWIN TECHNOLOGY IN POWER SYSTEMS



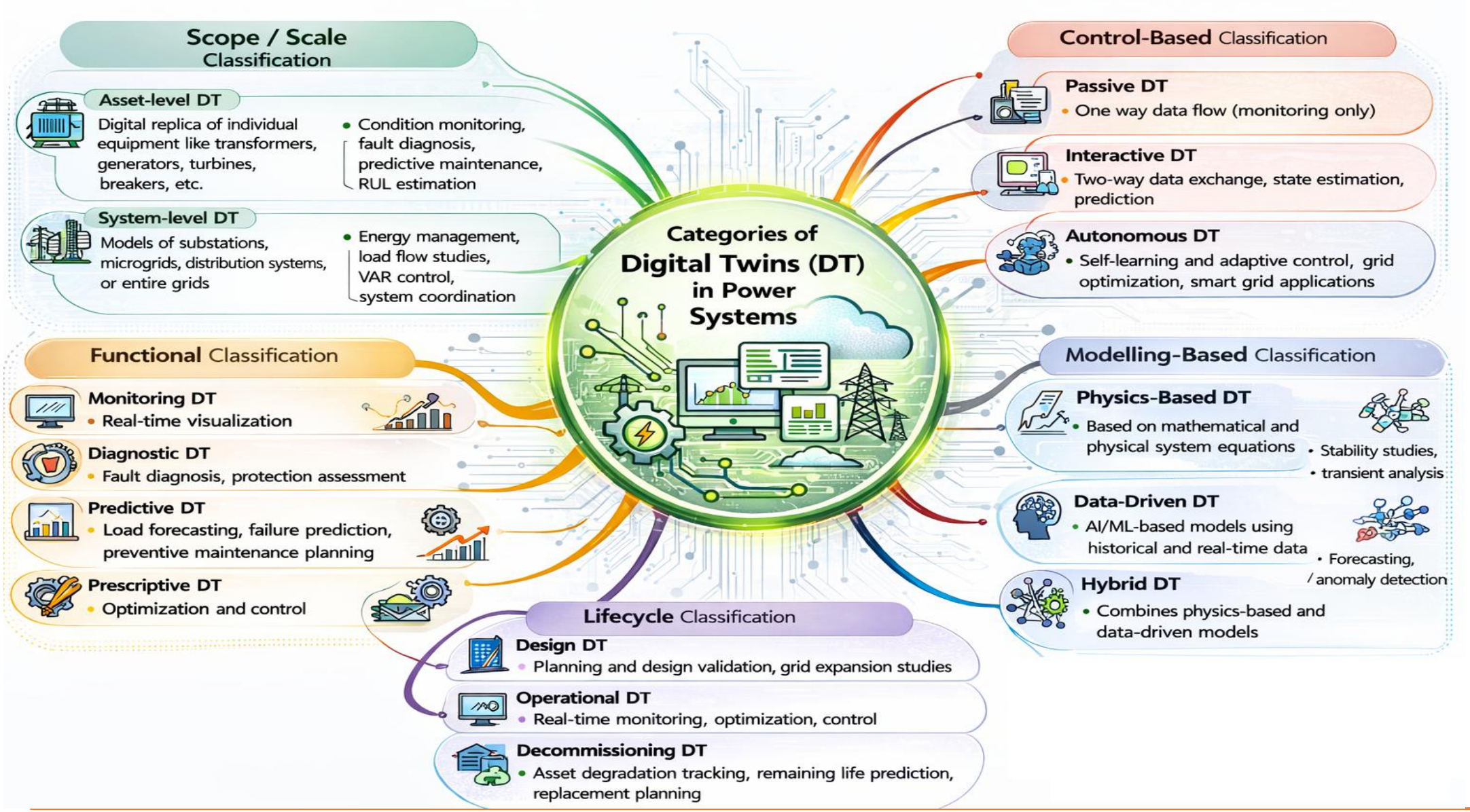
DIGITAL TWIN TECHNOLOGY IN POWER SYSTEMS

- Dynamic virtual replica of the physical grid
  - Uses real-time data (IoT , AI , physics-based models)
  - Continuously synchronizes with physical system
  - Enables simulation before real-world execution
- Predictive maintenance & fault prevention
- Real-time monitoring & load balancing
- Accurate forecasting & scenario simulation
- Operational optimization & reliability improvement
- Active decision-making (beyond static simulation models)

DT- SUPPORTING TECHNOLOGIES



DIGITAL TWIN FOR POWER GRID MODERNIZATION



CASE STUDIES

NREL (USA) – Control Room Digital Twin

- Integrated Digital Twin & AI/ML platform
- Enhances real-time grid control room operations
- Improves decision-making, optimization, and reliability
- Focus: **System-level operational intelligence**

TERI (India) – Cloud-Based Distribution DT

- Cloud-hosted DT for urban distribution networks
- Integrates smart meters, feeder models, and weather data
- Provides real-time visibility and analytics
- Focus: **Distribution grid modernization**

Siemens (Global) – Electrical Digital Twin

- "Single source of truth" for utilities
- Combines network modelling + predictive simulations + real-time insights
- Supports planning, operability studies, and construction
- Focus: **Transmission & distribution planning and operations**

GE Vernova (Global) – Asset-Level Digital Twins

- ML-enabled DT for turbines and generators
- Uses SCADA/OT data + physics models + anomaly detection
- Reduces forced outages and maintenance costs
- Focus: **Generation asset optimization**

DT TECHNOLOGY - POWER SECTOR – INDIA

DT TECHNOLOGY - POWER SECTOR – INDIA

Current Status –DT Technology:

- In early adoption stage, mainly through pilot projects and research initiatives.
- DT platforms for monitoring substations, transformers, feeders, and smart meters.
- Digital twins are being explored under Smart Grid initiatives and digital infrastructure programs.
- Digital twins are increasingly linked with India's Smart Grid Mission and smart meter rollout programs.
- Research institutions and universities in India are developing digital twin models for power grids and renewable energy systems.
- Current applications include real-time grid monitoring, predictive maintenance, load forecasting, and fault detection.

Challenges in Implementation in India:

- Data Availability and Quality
- High Initial Investment
- Lack of Skilled Workforce
- Interoperability Issues
- Cybersecurity Concerns

Future Direction of Digital Twins in India's Power Sector

- AI-powered digital twin of the national power grid to stabilize operations and optimize energy flows.
- Integration with Renewable Energy
- Smart Grid and Smart City Integration
- AI-driven Autonomous Grid Operation
- Edge Computing and Cloud Platforms
- Digital Twin-based Energy Planning

SUMMARY

- ✓ Digital Twins are transforming conventional grids into intelligent, adaptive power systems.
- ✓ They enable real-time insight, predictive intelligence, and optimized decision-making.
- ✓ Integration of AI and advanced analytics strengthens grid resilience and reliability.
- ✓ Despite existing challenges, rapid digital advancements are accelerating adoption.
- ✓ Digital Twins will be a cornerstone of sustainable, future-ready power systems.

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